VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Module Bank

Module 2
Academic Year 2025-26

Staff Name: Dr. Sreekar Guddeti

Program Name: B. Tech
Branch: AI/ML
Year: 1
Semester: 1
Course: EP
Code: 25PY101

Section number: 28, 34

Instructions:

1. Answer all questions.

1. Quantum Free Electron Theory (QFET)

a) Compare and contrast the postulates of classical free electron theory with that of quantum free electron theory. [BT 1]
b) What are the merits of quantum free electron theory? [BT 1]
c) What are the demerits of quantum free electron theory? [BT 1]

2. Fermi-Dirac Distribution

a) Plot the probability of occupancy of energy level in the case of Maxwell-Boltzmann distribution and Fermi-Dirac distribution at
   i. $T = 0K$,

ii. $T \neq 0K$, and

iii. $T = \infty\, K$ [BT 2]

b) Five free electrons exist in a three dimensional infinite potential well of with all three widths equal to $a = 12$ Å.

i. Determine the Fermi energy level at $T = 0K$.

ii. Repeat part (i) for 13 electrons. [BT 3]

c) Fermi level of a metal is 6.25 eV and electrons in this material follow the Fermi-Dirac distribution. Calculate the temperature at which there is 1% probability that a state 0.30 eV below the Fermi energy level will **not** contain an electron. [BT 4]

3. Electronic specific heat of solids

a) Define specific heat of a material. [BT 1]

b) What is the expression for specific heat capacity from classical free electron theory. [BT 1]

c) Derive the expression for specific heat capacity from classical free electron theory and quantum free electron theory. [BT 2]

4. Density of states

a) What is the meaning of density of states function? [BT 1]

b) If the density of states of a material is a constant $Z(E) = k$ and the electron density is n, what is the Fermi level at $T = 0K$? [BT 3]

c) Sketch the density of states function

i. for the conduction band of a semiconductor.

ii. Repeat part (i) for the valence band. [BT 2]

5. E-k diagram

a) Sketch the $E\ vs\ k$ for a free electron. [BT 1]

b) Sketch the $E\ vs\ k$ for an electron in a periodic potential. [BT 1]

c) Consider a semiconductor with all electrons occupying states in the lowest valence band at $T = 0K$. Plot the $E\ vs\ k$ showing occupancy of levels at both $T = 0K$ and $T \neq 0K$. [BT 2]

6. Fermi level in semiconductors

a) Sketch a diagram showing the density of states, Fermi-Dirac probability function and areas representing electron and hole concentration for an intrinsic semiconductor. [BT 1, 2]

b) Repeat part (a) for the case of

i. n-type semiconductor, and

ii. p-type semiconductor. [BT 1, 2]

c) **Design** an n-type Si so that the Fermi level is 0.20 eV below the conduction band edge. [BT 3]

7. p-n junction diode

a) Define the built-in potential voltage and describe how it maintains thermal equilibrium. [BT 1]

b) Consider a silicon pn junction at $T = 300K$ with doping concentrations of $N_a = 2 \cdot 10^{17} cm^{-3}$ and $N_d = 10^{15} cm^{-3}$. Calculate the built-in potential barrier. [BT 2]

c) Discuss the change in width of depletion region

i. at forward bias, and
ii. at reverse bias. [BT 2]

8. Solar cell

a) Sketch the $I - V$ characteristics of pn junction solar cell. [BT 1]
b) Define short circuit current and open circuit voltage. [BT 1]
c) The energy band gap of a semiconductor is $3\ eV$. Is this material useful for solar cell applications? Give reasons. [BT 4]

9. Direct and indirect band gap semiconductors

a) Sketch the $E\ vs\ k$ diagram for
   i. a direct bandgap semiconductor, and
   ii. an indirect bandgap semiconductor. [BT 1]
b) Why is direct band gap semiconductor useful for solar cell application? Crystalline silicon is indirect band gap semiconductor. What is the process employed in industry to “convert” it into a direct band gap semiconductor for solar cell application? [BT 2]
c) Which material is used in blue LED? If the wavelength of blue photon is $0.4\ \mu m$, what is the bandgap of the material? [BT 3]

10. LED

a) What are the materials used for red, green, and blue LEDs? [BT 1]
b) What are the mechanisms that decrease the efficiency of an LED? How does heterojunction-based LED address the efficiency problem? [BT 1]
c) Calculate the critical angle at GaAs - air interface. The refractive index of GaAs is 3.8 at $\lambda = 0.70\ \mu m$. [BT 3]

End of module bank